

Dr. JAYANTA PAUL

Resume

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Google Scholar
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Education

- 2025 **PhD, Computer Science & Engineering**, *Indian Institute of Engineering Science and Technology (IIST)*, Shibpur (Formerly Bengal Engineering and Science University, Shibpur), *Specialization on: Machine Learning, Deep Learning, Computer Vision, Natural language processing.*
Thesis Title: Deep Learning-Driven Framework for Multimodal Social Media Content Analysis;
Thesis Defense Completed on: 27.11.2025, Date of Award: 19.12.2025
- 2020 **Master of Technology, Computer Science & Engineering**, *Indian Institute of Engineering Science & Technology*, Shibpur (Formerly Bengal Engineering and Science University, Shibpur).
- 2017 **Bachelor of Technology, Computer Science & Engineering**, *Maulana Abul Kalam Azad University of Technology, West Bengal (MAKAUT)*, (Formerly West Bengal University of Technology).

Work Experience

- July 2026 – **Institute Post-Doctoral Fellow**, *Department of Artificial Intelligence, Indian Institute of Technology, Kharagpur, West Bengal, India - 721302..*
Present
- April 2026 – **CHANAKYA Postdoctoral Fellow**, *AI4ICPS I Hub Foundation and Department of Artificial Intelligence, Indian Institute of Technology, Kharagpur, West Bengal, India - 721302..*
July 2026
- August 2025 – **Assistant Professor**, *Department of Computer Application and Science, Institute of Engineering and Management (IEM), Kolkata, 700091, Ashram Building, Sector-V, Salt Lake Electronics Complex.*
April 2026

Publications

Journal Articles

- 2026 Abhijit Mitra, Souvik Mondal, Anubhav Bhattacharya, **Jayanta Paul**, and Jaya Sil., Mask-guided rationales meet LLMs: Hybrid encoder–LLM question answering framework for grounded, faithfulness-regularized explainable hate speech detection., In ***Applied Soft Computing*, volume 200, article 115373. Elsevier.**
- 2026 Abhijit Mitra, Rajdeep Majumdar, Rivujit Das, Arghyadeep Saha, **Jayanta Paul**, Souvik Mondal, and Jaya Sil., Breaking Bias: A Context-Aware Multimodal Framework for Detecting and Neutralizing Ideological Bias in News., In ***ACM Transactions on Knowledge Discovery from Data*, volume 20, issue 3, article 38, pages 1–69. ACM.**
- 2025 **Jayanta Paul**, Anuska Roy, Abhijit Mitra, and Jaya Sil., HyV-Summ: Social media video summarization on custom dataset using hybrid techniques., In ***Neurocomputing*, volume 614, article 128852. Elsevier.**
- 2025 Abhijit Mitra, **Jayanta Paul**, Tanis Ahamed, Sagar Basak, and Jaya Sil., From Words to Visuals: Bridging Text and Visual Insights Using MetA-MARC Framework for Enhanced Scholarly Article Categorization., In ***Knowledge-Based Systems*, volume 324, article 113896. Elsevier.**
- 2025 **Jayanta Paul**, Siddhartha Mallick, Abhijit Mitra, Anuska Roy, and Jaya Sil., Multi-Modal Twitter Data Analysis for Identifying Offensive Posts Using a Deep Cross-Attention-Based Transformer Framework., In ***ACM Transactions on Knowledge Discovery from Data*, volume 19, issue 3, article 66, pages 1–30. ACM.**

- 2025 **Jayanta Paul**, Abhijit Mitra, Divyanshu Kulhadiya, Tejas Pawar, Angshuman Roy, and Jaya Sil., A Comprehensive Approach to Real-Time Attendance Systems: Integrating Face Recognition and Emotion Detection with Web Technologies., In ***Procedia Computer Science*, volume 258, pages 3436–3446. Elsevier.**
- 2024 **Jayanta Paul**, Ahel Das Chatterjee, Devtanu Misra, Sounak Majumder, Sayak Rana, Malay Gain, Anish De, Siddhartha Mallick, and Jaya Sil., A Survey and Comparative Study on Negative Sentiment Analysis in Social Media Data., In ***Multimedia Tools and Applications*, volume 83, issue 30, pages 75243–75292. Springer.**
- 2023 Siddhartha Mallick, **Jayanta Paul**, and Jaya Sil., Response Fusion Attention U-ConvNext for Accurate Segmentation of Optic Disc and Optic Cup., In ***Neurocomputing*, volume 559, article 126798. Elsevier.**
- 2023 Rajat Subhra Bhowmick, Rahul Indra, Isha Ganguli, **Jayanta Paul**, and Jaya Sil., Breaking CAPTCHA System with Minimal Exertion through Deep Learning: Real-Time Risk Assessment on Indian Government Websites., In ***Digital Threats: Research and Practice*, volume 4, issue 2, article 28, pages 1–24. ACM.**
- 2023 Rajat Subhra Bhowmick, Isha Ganguli, Ananya Paul, **Jayanta Paul**, and Jaya Sil., Improving Indic Code-Mixed to Monolingual Translation Using Mixed Script Augmentation, Generation & Transfer Learning., In ***ACM Transactions on Asian and Low-Resource Language Information Processing. ACM.***
- 2022 Rajat Subhra Bhowmick, Isha Ganguli, **Jayanta Paul**, and Jaya Sil., A Multimodal Deep Framework for Derogatory Social Media Post Identification of a Recognized Person., In ***ACM Transactions on Asian and Low-Resource Language Information Processing*, volume 21, issue 1, article 2, pages 1–19. ACM.**

[Manuscripts Under Review / Communicated](#)

- 2025 **Jayanta Paul**, Abhijit Mitra, and Jaya Sil., IndiToon MemeNet: A Deep Learning Framework for Unethical Content Detection and Person Recognition in Indian Cartoon Memes., *Information Fusion, Elsevier* – Communicated.
- 2025 Siddhartha Mallick, Aayushman Ghosh, **Jayanta Paul**, and Jaya Sil., CATFA-Net: A Trans-Convolutional Approach for Accurate Medical Image Segmentation., *Medical Image Analysis, Elsevier* – Under revision.

[Conference and Book Chapter](#)

- 2025 **Jayanta Paul**, Sanju De, Angshuman Roy, Arkapravo Das, and Abhijit Mitra, Deep Learning-Powered IoT System for Real-Time Environmental Health Risk Monitoring and Proactive Alerts, In ***7th International Symposium on Devices, Circuits and Systems May, 2025*** .
- 2024 **Jayanta Paul**, *Abhijit Mitra*, Somak Sanyal, and Jaya Sil., Socialfaceemonet: A deep architecture for social media face recognition and emotion detection using customized datasets., In ***International Conference on Pattern Recognition*, pages 309–327. Springer.**
- 2024 Abhijit Mitra, Rivujit Das, **Jayanta Paul**, Rajdeep Majumder, Arghyadeep Saha, Nandita Sengupta, and Jaya Sil., A deep learning framework for classifying and mitigating bias in news reporting., In ***Proceedings of the Future Technologies Conference*, pages 424–443. Springer.**
- 2023 Siddhartha Mallick, Nitu Saha, **Jayanta Paul**, Isha Ganguli, Shantonu Debnath, and Jaya Sil., An efficient deep learning framework for glaucoma diagnosis using convolution mixed transformer network. , In ***TENCON 2023-2023 IEEE Region 10 Conference (TENCON)*, pages 1111–1116. IEEE.**
- 2022 **Jayanta Paul**, Anuska Roy, Siddhartha Mallick, and Jaya Sil. , A comparative study of deep learning-based face recognition and emotion detection techniques using social media customized cartoon post. , In ***International Conference on Computational Intelligence in Pattern Recognition*, pages 401–411.**

- 2022 **Jayanta Paul**, Rajat Subhra Bhowmick, and Jaya Sil. , Low-computation iot system framework for face recognition using deep learning algorithm. , In **International Conference on Computational Intelligence in Pattern Recognition**, pages 24–35. Springer.
- 2022 Siddhartha Mallick, **Jayanta Paul**, Nandita Sengupta, and Jaya Sil. , Study of different transformer based networks for glaucoma detection. , In **TENCON 2022-2022 IEEE Region 10 Conference (TENCON)**, pages 1–6. IEEE.
- 2021 **Jayanta Paul**, Rajat Subhra Bhowmick, Baisakhi Das, and Biplab K Sikdar. , A smart home security system in low computing iot environment. , In **IEEE 17th India council international conference (INDICON)**, pages 1–7. IEEE.
- 2020 **Jayanta Paul**, Rajat S Bhowmick, Riom Sen, Dwaipayan Ray, Suman S Manjhi, Soumya Sen, and Biplab K Sikdar. , Evaluation of face recognition schemes for low-computation iot system design, In **24th International Symposium on VLSI Design and Test (VDAT)**, pages 1–6. IEEE.

Fellowships & Awards

- 2026 – 2027 **Chanakya Fellowship** CHANAKYA Postdoctoral Fellow at the Department Artificial Intelligence and AI4ICPS Innovation Hub, IIT Kharagpur.
- 2020 – 2025 **Institute Fellowship** as a Ph.D. research scholar at Indian Institute of Engineering Science and Technology (IIST), Shibpur.
- 2017 Achieved 1st position making quadcopter (robotics championship zonal round) under ARK Techno solutions and robokart.com (IIT Madras).

Academic Achievements & Recognitions

- 2026 Invited to take the lecture on **“Mobile & Digital Trace Data Using Mobility Patterns for Urban Planning”** during the Residential Training Programme on Big Data, AI, and ML using Python for the ISS Probationers funded by the **Ministry of Statistics and Programme Implementation (MoSPI), Govt of India and organized by the IIT Kharagpur AI4ICPS I Hub Foundation at IIT Kharagpur** on 31st May, 2026.
- 2023 Invited for a talk in **“The Innovative Power of Machine Learning in Image Processing”**, during 6th-10th November, 2023 at **Faculty Development Programme Organization- MCKV Institute of Engineering, Lilua, Howrah.**
- 2017 Qualified GATE Examination

Technical skills

- Programming Languages Python, PyTorch, TensorFlow, C
- Subject Machine learning, Deep learning, Computer Vision.

Position of Responsibility

- 2014-2017 **Class representative in CSE Department**, Global Institute Of Management And Technology.
- 2023-2024 **Year representative**, Ph.D., CST Department, Indian Institute of Engineering Science and Technology (IIST).
- 2023-2024 **Treasurer**, *Academic Society of Computer Engineers*, CST Department, Indian Institute of Engineering Science and Technology (IIST).
- March 22-24, 2023 **Organizer, Revelation**, *The Departmental Fest of CST* , Indian Institute of Engineering Science and Technology (IIST).
- April 4-6, 2024 **Committee Convenors, Revelation**, *The Departmental Fest of CST* , Indian Institute of Engineering Science and Technology (IIST).

March 21-23, **ML Event Lead, Revelation**, *Academic Society of Computer Engineers*, CST Department,
2025 Indian Institute of Engineering Science and Technology (IIST).

Laboratory Assistantship

- 2022-2024 : **CS4271: Artificial Intelligence Lab**, Indian Institute of Engineering Science and Technology (IIST).
- 2021-2023 : **CS4172: Machine Learning Laboratory**, Indian Institute of Engineering Science and Technology (IIST).
- 2019-2024 : **CS1271: Introduction to Computing**, Indian Institute of Engineering Science and Technology (IIST).
- 2022-2024 : **CS3273: Software Engineering Lab**, Indian Institute of Engineering Science and Technology (IIST).
- 2021-2024 : **CS3183: Soft Computing Lab**, Indian Institute of Engineering Science and Technology (IIST).

References

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Personal Information

Father : Mr. Ranjan Kumar Paul
Gender : Male
Languages : Bengali, Hindi and English
Hobby : Playing cricket, playing guitar, teaching, editing videos
DOB : 10.02.1995